

Practical : 7	Implementation of making a change problem using dynamic programming.
Date :	

Objective :- To implement the Making Change Problem using Dynamic Programming and determine the minimum number of coins required to make a given amount.

Program :-

```
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;

int minCoins(vector<int> coins, int amount)
{
    const int INF = 1000000;

    // DP array
    vector<int> dp(amount + 1, INF);

    // Base case
    dp[0] = 0;

    // Calculate minimum coins for every amount
    for (int i = 1; i <= amount; i++)
    {
        for (int coin : coins)
        {
            if (coin <= i && dp[i - coin] != INF)
            {
                dp[i] = min(dp[i], dp[i - coin] + 1);
            }
        }
    }

    if (dp[amount] == INF)
        return -1;

    return dp[amount];
}
```

```
int main()
{
    vector<int> coins = {1, 2, 5};
    int amount = 11;

    int result = minCoins(coins, amount);

    if (result == -1)
        cout << "Amount cannot be formed." << endl;
    else
        cout << "Minimum number of coins = " << result << endl;

    return 0;
}
```

```
Minimum number of coins = 3
```

Analysis:-

Let:

- **n = number of different coin denominations**
- **A = target amount**

The program has two nested loops:

```
for (int i = 1; i <= amount; i++)
```

```
{
```

```
    for (int coin : coins)
```

Outer loop

Runs A times.

$O(A)$

Inner loop

For every amount, we check all n coins.

$$O(n)$$

Therefore:

$$\boxed{\text{Time Complexity} = O(A \times n)}$$

Example

If:

- Amount = 100
- Number of coins = 5

Then approximately:

$$100 \times 5 = 500$$

coin checks are performed.

Conclusion:-

The Making Change Problem was successfully implemented using Dynamic Programming. The algorithm calculates the minimum number of coins required to form a given amount by storing solutions to smaller subproblems. For coins = {1, 2, 5} and amount = 11, the minimum number of coins required is 3 (5 + 5 + 1). The time complexity is $O(A \times n)$ and the space complexity is $O(A)$.